

ASSESSMENT

Thank you for your interest in joining **Expert Thermal / XThermal**. This assessment is designed to evaluate your AI/ML fundamentals, ability to work with engineering calculation code, understanding of physics-informed AI, and ability to think about AI integration in an engineering software platform.

- 1. Problem Statement:** You are provided with a Python heat-sink calculation script. The model estimates thermal resistance and junction temperature based on heat load, airflow, TIM properties, and heat-sink parameters. Your task is to use the provided code as a base model, generate data through parameter sweeps, build a simple ML surrogate model, and explain how AI/ML can be applied to thermal engineering workflows.

Task 1: Python Model, Parameter Sweep, and ML Surrogate Model

Using the provided heat-sink calculation code, modify the model to vary the following parameters:

TDP: 30 W to 250 W

Air velocity: 0.5 m/s to 15 m/s

TIM thermal conductivity, k_tim: choose a reasonable range, such as 1–12 W/m·K

Create a dataset from the parameter sweep and save it as a CSV file.

Then build a simple ML surrogate model to predict:

- Total thermal resistance
- Junction temperature

Use at least two relevant ML models: Justification and evaluation of the 2 models.

Evaluate model accuracy using:

- MAE
- RMSE
- R^2 score

Also, perform a basic correlation or sensitivity analysis to identify which parameters have the strongest influence on thermal resistance and junction temperature.

Task 2: Agentic AI / Workflow AI Understanding

In your own words, (No AI assistance) Please explain your understanding of:

- Agentic AI
- Workflow AI
- AI copilots / assistants
- Difference between a standalone AI model and AI integrated into a product workflow

Also describe any past experience you have with:

- GenAI, LLMs, RAG, AI agents
- Workflow automation
- AI integration into a web platform or product

If you do not have direct experience, explain how you would approach learning and implementing these concepts.

Task 3: AI Use Case Evaluation

Suggest 3–5 AI features that could be useful in an engineering software platform.

For each feature, briefly mention:

- What problem it solves;

ASSESSMENT

- What data would be required;
- Whether it requires ML, GenAI, workflow AI, or a combination;
- How you would validate that it works correctly.

Examples of possible areas include:

- AI-assisted input validation
- Missing parameter detection
- Result summarization
- Thermal design recommendations
- Report generation
- Knowledge retrieval from engineering documents
- ML surrogate models for fast prediction
- Physics-aware AI for future thermal analysis

Task 4: Motivation and Ownership

Please answer the following(In your own words):

- Why do you want to join Expert Thermal / XThermal?
- What are you most interested in building in AI/ML?
- Are you more interested in ML models, GenAI workflows, physics-aware AI, or product AI integration?
- How do you handle ambiguous problems where requirements are not fully defined?
- Are you comfortable working in a fast-moving startup environment?
- How do you think you can grow within our team?
- This role calls for ownership, commitment, and initiative. How would you bring new ideas, create solutions, and take on greater responsibility over time?

Deliverables:

- a. Python script or Jupyter notebook for the modified heat-sink model and parameter sweep.
- b. Generated CSV dataset.
- c. ML surrogate model implementation and accuracy results.
- d. Correlation / sensitivity analysis.
- e. Written responses for Tasks 2–5.
- f. A concise summary report including:
 - Assumptions and their justifications
 - Key results from the model
 - ML model accuracy comparison
 - Important correlations or sensitivities
 - Limitations of the approach
 - Areas for improvement or optimization
 - You may also include GitHub, portfolio, or demo links if available.

Submission Deadline: 2 days from the day of receipt.